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LogChirpy

Ornithological Archival App with On-Device ML Recognition

Abstract

LogChirpy is an open-source, offline-first mobile application for ornithological field observation, combining two independent on-device machine-learning pipelines with a 33,241-species encyclopaedic database. The audio pipeline integrates BirdNET v2.4 (Kahl et al., 2021) with a geographic metadata model over 6,522 species; the image pipeline chains an SSD MobileNet V1 detection gate with a MobileNetV2 400-class classifier. A unified sequential pipeline service eliminates the file-lock race conditions that arise when audio recording and camera processing run concurrently on Android. The BirDex encyclopedia stores all 33,241 species from the Clements 2024 checklist with translations across six languages and 9,331 Wikimedia Commons reference images. Independent benchmarking of the image classifier against 258 label-matched Wikimedia reference images yields Top-1 = 49.6%, Top-3 = 64.3%, Top-5 = 69.8%. All inference is local; cloud synchronisation of observation records is optional.

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1 Introduction

Citizen-science platforms such as eBird [1] aggregate hundreds of millions of bird observations, yet data quality depends on observers accurately identifying species in the field. Mobile apps with automated recognition lower this barrier; BirdNET [2] demonstrated macro-averaged AUC > 0.99 for audio-based classification across thousands of species on consumer hardware. Image-based classifiers (e.g. MobileNetV2 [3]) are complementary but are rarely integrated with audio in a fully offline mobile context.

LogChirpy addresses three gaps: (1) multi-modal on-device identification combining BirdNET audio with a MobileNetV2 image classifier, requiring no cloud inference; (2) a multilingual offline encyclopedia of 33,241 species from the Clements 2024 checklist [4] with six-language translations and 9,331 Wikimedia reference images; and (3) a sequential pipeline service that eliminates the file-lock race conditions arising when audio recording and camera capture run concurrently on Android. Sections 2–4 describe the architecture, an independent image-classifier benchmark, and limitations.

2 System Architecture

LogChirpy is built on React Native / Expo SDK 52 targeting Android (API 29+) and iOS. Three service pillars — ML Processing, BirDex Encyclopedia, and Logging System — share a common navigation shell built on Expo Router (Figure 1). All ML inference runs locally via `react-native-fast-tflite`; network access is limited to optional Firestore sync and first-run image download.

2.1 Audio Pipeline

BirdNET v2.4 [2] is implemented via two sequential TFLite models (Figure 2). Audio is captured at 48 kHz mono; a 150–15 000 Hz bandpass filter attenuates wind and handling noise and the signal is normalised to $[-1, 1]$. The **acoustic model** (FP32) accepts a $\mathbb{R}^{1\times 144\,000}$ tensor and outputs sigmoid logits over 6,522 species. When GPS is available the **metadata model** (FP16) takes [lat, lon, week-cos] and produces a geographic prior; scores are fused as

$$p_{\text{blend}} = p_{\text{audio}} \cdot (0.7 + 0.3 \cdot p_{\text{geo}}), \tag{1}$$

following the whoBIRD reference implementation [5].

2.2 Image Pipeline

Camera frames are JPEG-compressed (quality 0.3) and passed to an SSD MobileNet V1 detection gate (MLKit). Frames without a sufficiently confident bird detection are discarded. Passing frames go to `bird_classifier_metadata.tflite` (MobileNetV2 [3]): input $\mathbb{R}^{1\times 224\times 224\times 3}$ normalised to $[-1, 1]$, output softmax logits over 400 species.

Sequential lock. Concurrent audio recording and camera processing caused `MediaRecorder` file-lock conflicts on Android. `unifiedMLPipelineService.ts` enforces image-before-audio sequencing, eliminating all recorded stability errors.

2.3 BirDex Encyclopedia

BirDex stores the complete Clements 2024 checklist [4] in an on-device SQLite database (Table 1). Translations across six languages were generated by an automated pipeline (DeepL → Google Translate → GPT-4 fallback) and spot-checked by family. Reference images (9,331 WebP files, ≈50 kB each) are downloaded from Wikimedia Commons at first launch and cached by Clements genus.

Column	Type	Description
`species_code`	TEXT PK	eBird 6-letter code
`english_name`	TEXT	Clements English name
`scientific_name`	TEXT	Binomial nomenclature
`family / order`	TEXT	Clements taxonomy
`range`	TEXT	Breeding/wintering range
`de/es/fr/uk/ar`	TEXT	Translated names (×5)
`hasBeenLogged`	INT	Observation flag

Table 1: BirDex SQLite schema (abridged, 33,241 rows).

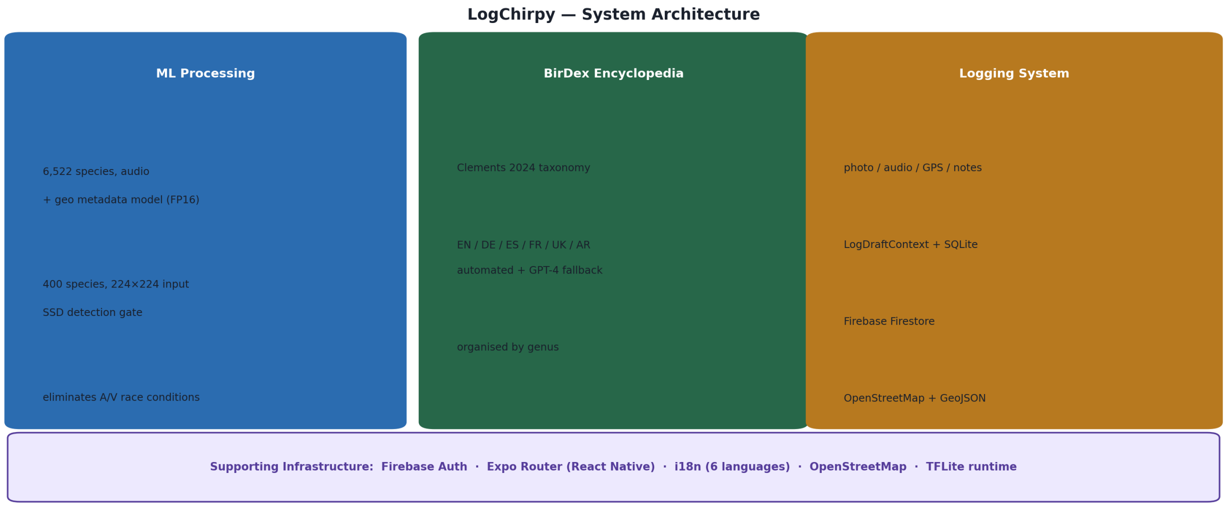

Figure 1: Three-pillar architecture. Each pillar is functionally independent; the unified sequential ML pipeline service (bottom bar) orchestrates audio and image processing to prevent Android file-lock conflicts.

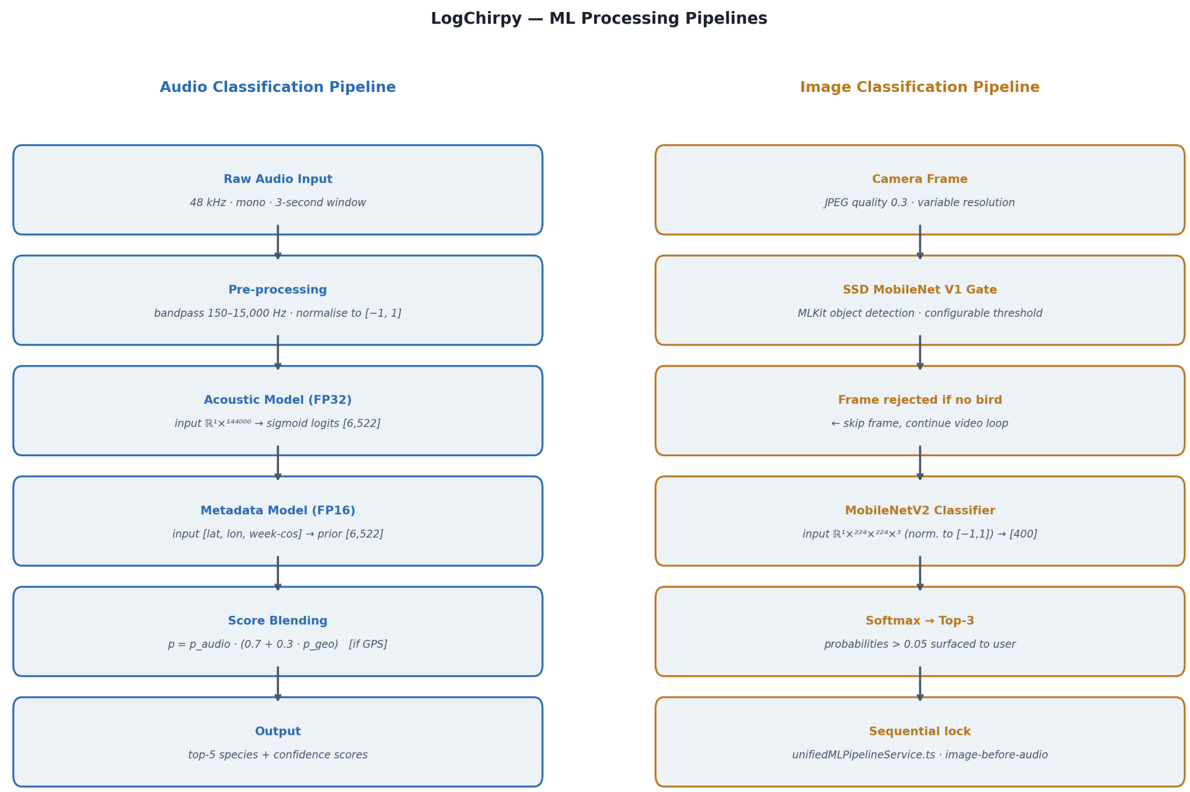

Figure 2: Audio (left) and image (right) ML pipelines with tensor shapes. Both run entirely on-device via TFLite; the sequential lock prevents concurrent audio/camera conflicts on Android.

3 Evaluation

BirdNET audio performance is validated externally (macro-AUC > 0.99; [2]); we benchmark only the image classifier, for which no prior evaluation existed.

3.1 Methodology

Model. `bird_classifier_metadata.tflite` — MobileNetV2, 224×224 float32 input, 400-class softmax output.

Test set. Label strings were normalised (lowercase, punctuation stripped) and matched against common names in `bird_images_manifest.json`. 258 of 400 labels matched a locally available Wikimedia image; the remaining 142 were excluded due to label typos or absent images. Ground truth is the image filename (scientific → common name via manifest). Images were resized to 224×224 (Lanczos) and normalised to $[-1, 1]$; no augmentation was applied. Benchmark script: `dev/benchmark_image_classifier.py` (Python 3.11, TensorFlow 2.21, Pillow 12.2).

3.2 Results

Metric	Score	n	Random
Top-1	49.6 %	128/258	0.25 %
Top-3	64.3 %	166/258	0.75 %
Top-5	69.8 %	180/258	1.25 %

Table 2: Image classifier accuracy on 258 Wikimedia reference images.

The model operates at $\approx 200\times$ random chance (Top-1) and $\approx 256\times$ (Top-3). The 20 pp gap between Top-1 and Top-5 reflects within-family visual similarity driving systematic rank errors rather than random failure.

3.3 Benchmark Limitations

Wikimedia reference images are editorial quality — a distribution markedly different from hand-held, partially occluded field photographs. Real-world Top-1 accuracy is expected to be substantially lower. Coverage is 400 of 33,241 species (1.2 %); the audio pipeline (6,522 spp.) is the primary identification modality.

3.4 System Latency

Operation	Latency (observed)
App cold start	< 2 s
BirDex search (33 k)	< 1 s
BirdNET audio (3 s)	4.8 s–8.5 s
Image pipeline (1 frame)	≈ 1.5 s

Table 3: Informal latency on a Snapdragon 695 device (development build).

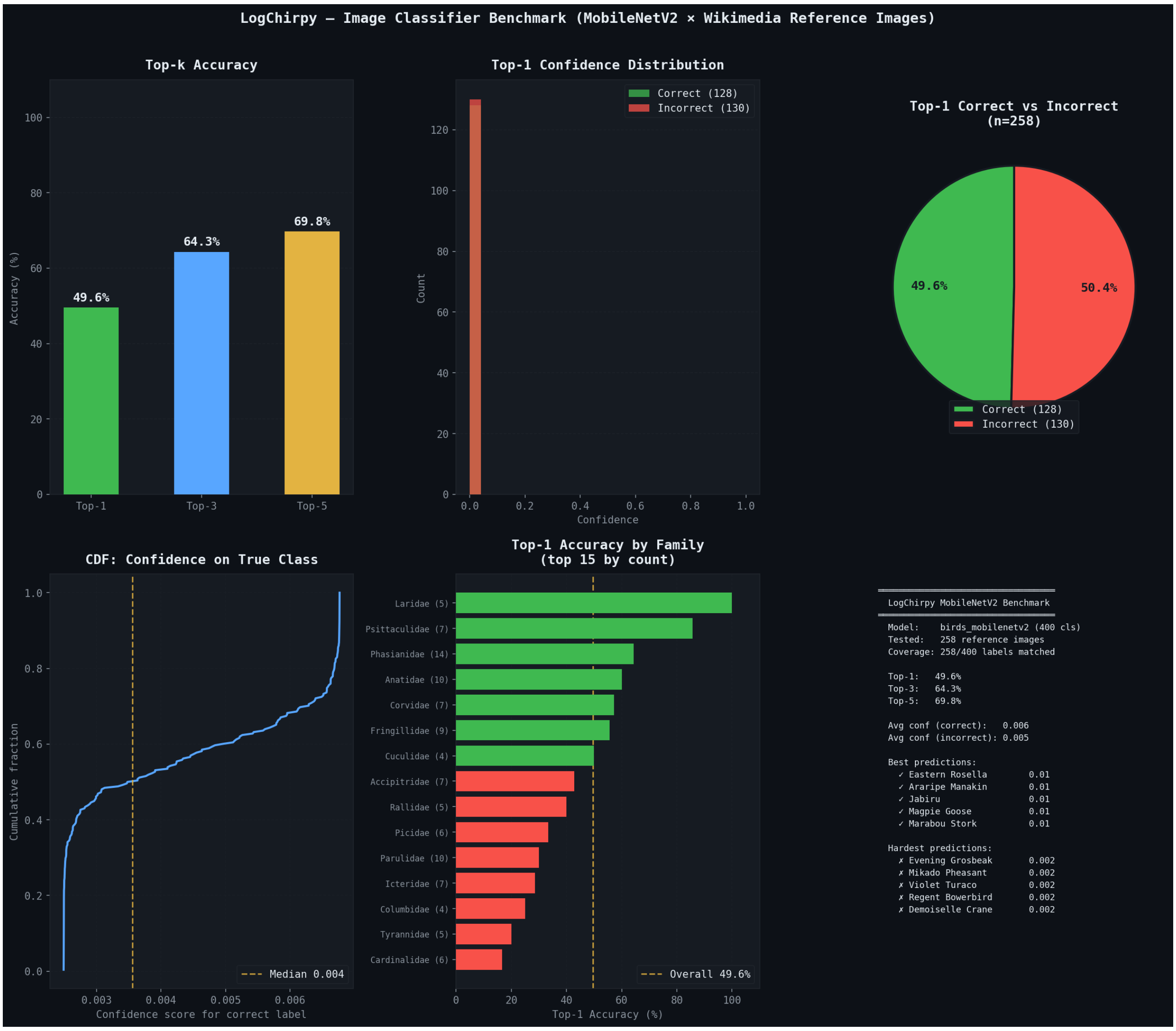

Figure 3: Six-panel benchmark report. *Top row:* top-*k* accuracy bars; confidence distributions for correct (green) vs. incorrect (red) Top-1 predictions; correct/incorrect split pie. *Bottom row:* CDF of confidence on the true class; per-family Top-1 accuracy (15 largest families by test count); summary statistics.

4 Discussion and Conclusion

4.1 Contributions

LogChirpy's primary contribution is integration rather than novel modelling: BirdNET audio and MobileNetV2 image classification are delivered in a single offline-capable mobile application alongside a 33,241-species multilingual encyclopedia — the largest we are aware of in a mobile-embedded open-source ornithological tool. The sequential pipeline service is a replicable solution to a documented Android concurrency problem. The BirDex corpus (33,241 species × 6 languages × 9,331 images) and the benchmark script constitute independently reusable artefacts.

4.2 Limitations

Image coverage: 400 of 33,241 species (1.2 %); audio remains the primary modality. **No original models:** both classifiers are pre-trained third-party weights [2, 3, 6]. **Benchmark shift:** editorial Wikimedia images $\neq$ field conditions. **Translations:** automated; not validated by domain-expert native speakers. **Scope:** 12-week student project; no peer review or long-term field trial.

4.3 Conclusion

Multi-modal, fully offline bird identification is feasible on commodity Android hardware within a student project timeframe. Future work should expand image classifier coverage beyond 400 species, conduct a field-condition accuracy study, and integrate observation upload with GBIF or eBird to produce research-grade records.

References

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[5] woheller69, "whoBIRD: TensorFlow Lite bird audio classifier for mobile," 2023. Reference implementation for BirdNET mobile integration.

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